

A symmetric sum that is a harmonic number

the partial fraction $\frac{1}{k(n-k)} = \frac{1}{n} \left(\frac{1}{k} + \frac{1}{n-k} \right)$ and Euler's constant

$$\lim_{n \rightarrow \infty} \left(\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} - \log n \right) = \gamma = 0.5772156649 \dots$$

Overview. The sum is not merely asymptotic to a harmonic number; it *equals* one. A single partial fraction turns $\frac{n}{2k(n-k)}$ into the average of $\frac{1}{k}$ and $\frac{1}{n-k}$, and the reflection $k \mapsto n-k$ makes the two halves identical, so

$$\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} = H_{n-1}, \quad H_m = \sum_{j=1}^m \frac{1}{j}.$$

The problem then reduces to the definition of the Euler–Mascheroni constant, $H_m - \log m \rightarrow \gamma$, with a harmless shift from $m = n-1$ to n .

The subtraction of $\log n$ is forced. Without it the sum diverges, and it diverges logarithmically because $\int_0^1 \frac{dx}{x(1-x)}$ diverges at both endpoints. Section 4 works this out.

1. The exact identity

For $1 \leq k \leq n-1$,

$$\frac{1}{k(n-k)} = \frac{k + (n-k)}{n k(n-k)} = \frac{1}{n} \left(\frac{1}{k} + \frac{1}{n-k} \right). \quad (1)$$

This is the only algebraic step. Multiplying by $\frac{n}{2}$ and summing,

$$\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} = \frac{1}{2} \sum_{k=1}^{n-1} \frac{1}{k} + \frac{1}{2} \sum_{k=1}^{n-1} \frac{1}{n-k}.$$

The substitution $j = n-k$ turns the second sum into the first:

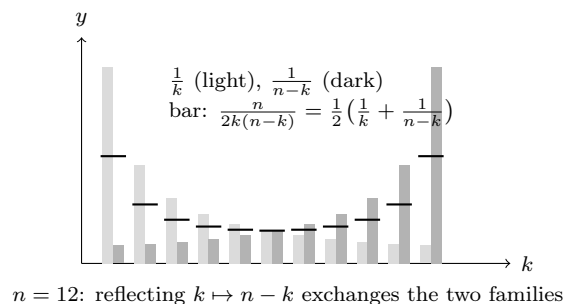
$$\sum_{k=1}^{n-1} \frac{1}{n-k} = \sum_{j=1}^{n-1} \frac{1}{j} = H_{n-1}.$$

Hence

$$\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} = H_{n-1} \quad \text{for every } n \geq 2. \quad (2)$$

No approximation has been made. For instance $n = 4$ gives $\frac{4}{2} \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{3} \right) = \frac{11}{6} = H_3$.

The factor $\frac{n}{2}$ in the statement of the problem is exactly the normalisation that makes (2) an identity, and the symmetry $k \leftrightarrow n-k$ is what supplies the 2.



2. Euler's constant

Proposition. The sequence $d_m = H_m - \log m$ converges.

Proof. Compare with the integral. Since $\frac{1}{x}$ is decreasing, for $j \geq 1$

$$\frac{1}{j+1} \leq \int_j^{j+1} \frac{dx}{x} \leq \frac{1}{j}. \quad (3)$$

Summing the right inequality for $j = 1, \dots, m-1$ gives $\log m \leq H_{m-1}$, so $d_m \geq \frac{1}{m} > 0$; summing the left gives $H_m - 1 \leq \log m$, so $d_m \leq 1$. For monotonicity,

$$d_{m+1} - d_m = \frac{1}{m+1} - \log \frac{m+1}{m} = \frac{1}{m+1} - \int_m^{m+1} \frac{dx}{x} \leq 0$$

by (3). A decreasing sequence bounded below converges. $\square$

Its limit is by definition the Euler–Mascheroni constant,

$$\gamma = \lim_{m \rightarrow \infty} (H_m - \log m) = 0.57721566490 \dots \quad (4)$$

3. Conclusion, with the index shift

By (2), the quantity in the problem is $H_{n-1} - \log n$. Write it in terms of the defining sequence:

$$H_{n-1} - \log n = [H_{n-1} - \log(n-1)] + \log \frac{n-1}{n} = d_{n-1} + \log \left(1 - \frac{1}{n}\right).$$

The first term tends to γ by (4) and the second to $\log 1 = 0$, so

$$\boxed{\lim_{n \rightarrow \infty} \left(\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} - \log n \right) = \gamma.}$$

Rate. Using the standard expansion $H_m = \log m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + O(m^{-4})$ with $m = n$ and $H_{n-1} = H_n - \frac{1}{n}$,

$$H_{n-1} - \log n = \gamma + \frac{1}{2n} - \frac{1}{12n^2} - \frac{1}{n} + O(n^{-4}) = \gamma - \frac{1}{2n} - \frac{1}{12n^2} + O(n^{-4}). \quad (5)$$

So the sequence approaches γ from below at rate $\frac{1}{2n}$, with the next correction $-\frac{1}{12n^2}$ visible in the table below.

n	$H_{n-1} - \log n$	$\gamma - \frac{1}{2n}$	difference
10	0.5263832	0.5272157	$-8.33 \cdot 10^{-4}$
100	0.5722073	0.5722157	$-8.33 \cdot 10^{-6}$
1000	0.5767156	0.5767157	$-8.33 \cdot 10^{-8}$

4. Why the answer had to involve γ

Rewrite the sum as a Riemann sum. With $x_k = \frac{k}{n}$,

$$\frac{n}{2} \sum_{k=1}^{n-1} \frac{1}{k(n-k)} = \frac{1}{2} \sum_{k=1}^{n-1} \frac{1}{n} \cdot \frac{1}{x_k(1-x_k)}. \quad (6)$$

The right-hand side is a Riemann sum for

$$\frac{1}{2} \int_0^1 \frac{dx}{x(1-x)},$$

which diverges logarithmically, and at both endpoints. That is a diagnosis, not a failure: a divergent Riemann sum whose integrand has simple poles at the endpoints produces a term $\log n$ plus a constant determined by the discrete structure, and the constant attached to $\frac{1}{x}$ sampled at $\frac{1}{n}, \frac{2}{n}, \dots$ is exactly γ .

Each endpoint contributes $\frac{1}{2} \log n + \frac{1}{2} \gamma$ by (1), and the two add up to $\log n + \gamma$. This is the structural reason the problem subtracts $\log n$ rather than, say, $\frac{1}{2} \log n$, and why the answer is γ rather than $\frac{\gamma}{2}$.

5. Variants

The same partial fraction handles a family of similar sums; the results follow from (2) and (4) with no new ideas.

Sum	Value or limit
$\sum_{k=1}^{n-1} \frac{1}{k(n-k)}$	$= \frac{2H_{n-1}}{n}$, exactly.
$\sum_{k=1}^{n-1} \frac{n}{k(n-k)} - 2 \log n$	$\rightarrow 2\gamma$.
$\frac{n^2}{2} \sum_{k=1}^{n-1} \frac{1}{k^2(n-k)}$	$= \frac{n}{2} \sum_{k=1}^{n-1} \left(\frac{1}{k^2} + \frac{1}{k(n-k)} \right) \rightarrow \infty$ like $\frac{\pi^2}{12} n$.
$\sum_{k=1}^{n-1} \frac{1}{k(n-k)} \binom{n}{k}^{-1}$	not reachable this way: the partial fraction (1) survives, but the binomial weight destroys the reflection-into-itself structure of the two halves.

The middle entry illustrates the mechanism cleanly. Applying (1) to $\frac{n^2}{2k^2(n-k)}$ gives $\frac{n}{2} \left(\frac{1}{k^2} + \frac{1}{k(n-k)} \right)$; the first part sums to $\frac{n}{2} \cdot \frac{\pi^2}{6} + o(n)$ and dominates, while the second contributes only $H_{n-1} = O(\log n)$. Whenever the partial fraction produces a convergent series, that series, not the harmonic one, sets the scale.